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# Data Cleaning Report

PowerPulse Phase 2: staging layer and data cleaning (project_staging)

### Result at a glance

| Measure | Value | Comment |
|---|---|---|
| Source rows landed (4 EIA-930 CSV files) | 87,720 | matches the manifest per file |
| Clean rows produced | 87,720 | 5 authorities x 17,544 UTC hours |
| Rows rejected | 0 | no invalid rows in the real data; rules proven by self-test |
| Duplicate (authority, UTC hour) keys | 0 | uniqueness enforced by a primary key |
| Blank cells kept as NULL in demand / forecast / total generation | 218 / 218 / 218 | on 316 authority-hours; never replaced by 0 |
| Negative net-generation values retained | 31,007 | real source data (e.g. storage charging) |
| Numeric fuel observations | 727,720 | equals the supplied baseline: 0 differences |

## 1. Scope and staging architecture

Phase 1 designed a warehouse of five dimensions and two hourly fact tables over real EIA-930 grid data for five balancing authorities (CISO, ERCO, ISNE, NYIS, PJM), 1 January 2024 to 31 December 2025. This step lands the four source CSV files in PostgreSQL and cleans them into one typed table. Python loads the files with COPY; SQL performs the cleaning inside the database, so every rule is visible, repeatable and logged.

**Source files > project_staging > project_dw > project_reporting**

| Staging table | Rows | Purpose |
|---|---|---|
| `stg_source_file` | 4 | Manifest: URL, SHA-256 and row count of each source file |
| `stg_authority_reference` | 5 | Five balancing authorities with names and regions |
| `stg_energy_source_mapping` | 19 | Nineteen fuel headers (original spelling) and their groups |
| `stg_eia930_balance_raw` | 87,720 | Landing table: every column TEXT, exactly as in the CSV |
| `stg_eia930_balance_clean` | 87,720 | Typed, standardised, de-duplicated; read by the dimension and fact loads |
| `stg_load_exception` | 0 | One row per rejected row or nulled value, with rule and reason |
| `stg_cleaning_log` | 65 | Before / after / affected counts for every rule |

Only the *reported* measures are staged: demand, demand forecast, total net generation and the 19 reported fuel columns. The imputed and adjusted variants published by EIA are deliberately not loaded, so reported and estimated values are never mixed. Fuel headers keep their original spelling in `stg_energy_source_mapping.source_column` so the fact load can map them exactly.

## 2. Source import and reconciliation (rule C01)

Each file was loaded with its own column list: the 2024 Jan-Jun file publishes 9 reported fuel columns, the other three publish 16. A fuel column that does not exist in a file is stored as NULL (structural absence), which is different from a blank cell inside an existing column. Loaded row counts were compared with `source_manifest.json`.

| Source file | Fuel cols | Manifest rows | Loaded rows | Status | SHA-256 (first 12) |
|---|---|---|---|---|---|
| EIA930_BALANCE_2024_Jan_Jun.csv | 9 | 21,835 | 21,835 | OK | 26768c495c3b |
| EIA930_BALANCE_2024_Jul_Dec.csv | 16 | 22,085 | 22,085 | OK | a602a8e577cf |
| EIA930_BALANCE_2025_Jan_Jun.csv | 16 | 21,715 | 21,715 | OK | fac4bb991dfa |
| EIA930_BALANCE_2025_Jul_Dec.csv | 16 | 22,085 | 22,085 | OK | 1146158b9724 |
| TOTAL | | 87,720 | 87,720 | OK | |

Source lineage is preserved: every row keeps its source file name, its row number in the full EIA file (`source_row`) and, in the clean table, the SHA-256 of the file it came from, so a fact row can be traced back to an immutable source snapshot.

# 3. Cleaning rules and before / after counts

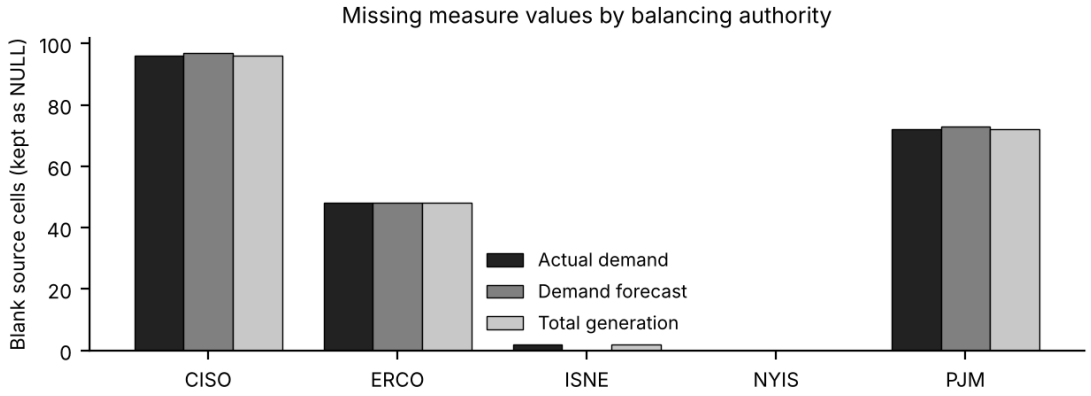
Each rule writes its counts to `stg_cleaning_log`. *Before* is the number of rows (or cells for C06-C08, C10) entering the rule, *after* the number leaving it, *affected* the number of rows or values the rule changed, rejected or flagged.

| Rule | Name | What it does | Before | After | Affected |
|---|---|---|---|---|---|
| C01 | Source import reconciliation | Compare rows loaded per file with the manifest; reject unknown file or invalid row number. | 87,720 | 87,720 | 0 |
| C02 | Text standardisation | TRIM and UPPER on authority and region codes (for example ' cISo ' becomes 'CISO'). | 87,720 | 87,720 | 0 |
| C03 | Reference validation | Authority must exist in the reference table and its region must agree with the reference. | 87,720 | 87,720 | 0 |
| C04 | Date and timestamp conversion | Parse Data Date to date, the local label to timestamp and the UTC label to timestamptz (+00:00). | 87,720 | 87,720 | 0 |
| C05 | Hour number and UTC offset validation | Hour Number must be 1-25; offset = local minus UTC must be -480, -420, -360, -300 or -240 minutes. | 87,720 | 87,720 | 0 |
| C06 | Numeric conversion | Convert text to numeric(18,4). A non-numeric value becomes NULL and is logged (cells). | 87,720 | 87,720 | 0 |
| C07 | NULL handling | Blank measure cells become NULL, never 0; the row is kept (cells). | 87,720 | 87,720 | 523,609 |
| C08 | Negative net generation retained | Negative net generation is real data: kept unchanged (cells). | 815,222 | 815,222 | 31,007 |
| C09 | Duplicate removal on | One row per (authority, UTC hour); newest snapshot wins; extra copies are logged as rejected. | 87,720 | 87,720 | 0 |
| C10 | Fuel mapping reconciliation | Numeric fuel cells per column compared with the supplied baseline (cells). | 727,720 | 727,720 | 0 |
| C11 | Final result | Raw rows versus clean rows plus rejected rows. | 87,720 | 87,720 | 0 |

Reading the table: C07 'affected' counts every blank cell (3 x 218 blank demand / forecast / generation cells plus blank fuel cells), all of which are stored as NULL. C08 'affected' counts negative values that were deliberately kept.

# 4. Handling missing values (rule C07)

Actual demand, demand forecast and total generation are each blank in **218** source rows. These fall on **316** distinct authority-hours, of which **120** are missing all three measures. Blank cells are stored as **NULL, never 0**, and the rows are kept: the hour still happened, only the measurement is unknown. Replacing a blank with 0 would pull averages and forecast-error metrics towards zero. A check confirms that the zeros in the clean table are exactly the zeros that exist in the raw data.




| Authority | Rows | NULL demand | NULL forecast | NULL total generation |
|---|---|---|---|---|
| CISO | 17,544 | 96 | 97 | 96 |
| ERCO | 17,544 | 48 | 48 | 48 |
| ISNE | 17,544 | 2 | 0 | 2 |
| NYIS | 17,544 | 0 | 0 | 0 |
| PJM | 17,544 | 72 | 73 | 72 |

### Example: blank source cells become NULL (raw text on the left, typed value on the right)

| Source row | BA | raw demand | raw forecast | raw generation | demand | forecast | generation |
|---|---|---|---|---|---|---|---|
| 2024_Jan_Jun / row 163243 | PJM | '' | '' | '' | NULL | NULL | NULL |
| 2024_Jan_Jun / row 163244 | PJM | '' | '' | '' | NULL | NULL | NULL |

**Structural absence:** the 2024 Jan-Jun file has no column for ten of the 19 fuel categories, and the other three files lack three. Those cells are NULL as well, and the clean table cannot tell them apart from a blank, so downstream steps must create generation facts only from numeric cells (blank and absent cells create no fact row). The C07 log entry records the number of structurally absent cells separately.

## 5. Type conversion and timestamps (rules C04, C05, C06)

Every raw column is TEXT. The clean table uses real types: `date`, `timestamp` (local wall-clock label), `timestamptz` (UTC instant) and `numeric(18,4)`. Timestamps are built with `make_timestamp`, so the server time zone can never shift them, and the UTC label is stored with an explicit +00:00. `utc_offset_minutes` is the local label minus the UTC instant.

| Row | raw date \| hour | raw local | raw UTC | date \| hour | local (timestamp) | UTC (timestamptz) | offset |
|---|---|---|---|---|---|---|---|
| 30571 | 01/01/2024 \| 1 | 01/01/2024 1:00:00 AM | 01/01/2024 9:00:00 AM | 2024-01-01 \| 1 | 2024-01-01 01:00:00 | 2024-01-01 09:00:00+00:00 | -480 |
| 163243 | 03/10/2024 \| 2 | 03/10/2024 3:00:00 AM | 03/10/2024 7:00:00 AM | 2024-03-10 \| 2 | 2024-03-10 03:00:00 | 2024-03-10 07:00:00+00:00 | -240 |
| 33943 | 11/03/2024 \| 25 | 11/04/2024 12:00:00 AM | 11/04/2024 8:00:00 AM | 2024-11-03 \| 25 | 2024-11-04 00:00:00 | 2024-11-04 08:00:00+00:00 | -480 |

**Spring clock change:** on 10 March 2024 the second hour of the day (Hour Number 2) carries the local label 03:00 because 02:00 does not exist; the offset changes from -300 to -240 minutes. **Autumn clock change:** Hour Number reaches 25 on the day clocks go back. The two repetitions of the same local label are different UTC instants, so both are kept (10 rows have hour number 25, one per authority per autumn). **Operating date:** the date comes from *Data Date*, never from the local label, because a midnight label belongs to the previous operating day (3,655 rows have a local-label date that differs from their operating date). **Offsets:** all rows fall in the five valid U.S. offsets (-480 to -240).

# 6. Negative net generation is retained (rule C08)

**31,007** numeric fuel observations are negative (storage charging, station load, solar at night). They are real source data, so they are kept as supplied. Total net generation has no negative values.

| Original fuel header | Clean column | Numeric cells | Negative |
|---|---|---|---|
| Net Generation (MW) from Unknown Energy Storage | gen_unknown_energy_storage | 9,985 | 9,985 |
| Net Generation (MW) from Other Fuel Sources | gen_other_fuel_sources | 87,473 | 9,303 |
| Net Generation (MW) from Solar without Integrated Battery Storage | gen_solar_without_integrated_battery_storage | 65,666 | 5,454 |
| Net Generation (MW) from Coal | gen_coal | 84,376 | 4,035 |
| Net Generation (MW) from All Petroleum Products | gen_all_petroleum_products | 67,529 | 969 |
| Net Generation (MW) from Solar | gen_solar | 21,807 | 951 |
| Net Generation (MW) from Battery Storage | gen_battery_storage | 20,497 | 266 |
| Net Generation (MW) from Hydropower Excluding Pumped Storage | gen_hydropower_excluding_pumped_storage | 65,666 | 44 |

| Source row | BA | raw text | clean value |
|---|---|---|---|
| 64572 | ERCO | -800 | -800.0000 |
| 64573 | ERCO | -1152 | -1152.0000 |
| 64574 | ERCO | -1055 | -1055.0000 |

# 7. Rejected and invalid records

In the real data **0 rows were rejected** and **0 values** were set to NULL because of bad content, so `stg_load_exception` is empty after a normal run. Because an empty exception table proves nothing about the rules, a self-test (`python etl/01_load_staging.py --self-test`) adds nine deliberately dirty rows to the raw table, runs the cleaning SQL, checks the outcome and then restores the real data.

| Rule | Reason | Action | Column | Bad value | Source row |
|---|---|---|---|---|---|
| C01 | UNKNOWN_SOURCE_FILE | ROW_REJECTED | source_file | NOT_IN_MANIFEST.csv | 900008 |
| C03 | UNKNOWN_AUTHORITY_CODE | ROW_REJECTED | balancing_authority | XXXX | 900003 |
| C03 | REGION_DOES_NOT_MATCH_AUTHORITY | ROW_REJECTED | region | TEX | 900006 |
| C04 | INVALID_DATA_DATE | ROW_REJECTED | data_date | 13/45/2043 | 900004 |
| C05 | INVALID_HOUR_NUMBER | ROW_REJECTED | hour_number | 26 | 900005 |
| C05 | INVALID_UTC_OFFSET | ROW_REJECTED | utc_offset_minutes | 0 | 900007 |
| C06 | NON_NUMERIC_MEASURE_SET_TO_NULL | VALUE_SET_NULL | demand_mw | N/A | 900009 |
| C09 | DUPLICATE_AUTHORITY_UTC_HOUR | ROW_REJECTED | utc_time_end | 01/01/2024 9:00:00 AM | 1 |

The two remaining dirty rows are correctly **kept**: ' pjm ' / 'mida ' was standardised to PJM / MIDA (C02), and a row whose demand was the text 'N/A' kept its generation values while demand became NULL (C06).

| Result | Self-test check |
|---|---|
| PASS | clean table = 87,720 real rows + 2 kept dirty rows |
| PASS | 7 dirty rows rejected, each with the expected rule and reason |
| PASS | 1 non-numeric cell set to NULL (rule C06) |
| PASS | ' pjm ' / 'mida ' standardised to PJM / MIDA (rule C02) |
| PASS | blank / non-numeric demand stays NULL, row kept |
| PASS | real data restored: 87,720 clean rows, 0 exceptions |

## 8. Duplicate removal (rule C09)

The business grain of the demand fact is one row per authority and UTC hour. The raw data contains **0** duplicate keys. The rule is nevertheless implemented: rows are ranked within each (authority_code, utc_hour_end) group by newest snapshot, then file name, then source row; rank 1 is kept and every other copy is written to the exception table. The primary key on the clean table makes a duplicate impossible, and the self-test proves the rule fires (source row 1 above was removed as a duplicate).

## 9. Fuel column reconciliation (rule C10)

Numeric observations per fuel column were compared with the baseline in `energy_source_mapping.csv`. All 19 columns agree; the total of **727,720** is the number of rows the fuel-generation fact table is expected to hold.

| Original fuel header (source_column) | Baseline | Clean | Status |
|---|---|---|---|
| Net Generation (MW) from All Petroleum Products | 67,529 | 67,529 | OK |
| Net Generation (MW) from Battery Storage | 20,497 | 20,497 | OK |
| Net Generation (MW) from Coal | 84,376 | 84,376 | OK |
| Net Generation (MW) from Geothermal | 384 | 384 | OK |
| Net Generation (MW) from Hydropower Excluding Pumped Storage | 65,666 | 65,666 | OK |
| Net Generation (MW) from Hydropower and Pumped Storage | 21,807 | 21,807 | OK |
| Net Generation (MW) from Natural Gas | 87,473 | 87,473 | OK |
| Net Generation (MW) from Nuclear | 87,472 | 87,472 | OK |
| Net Generation (MW) from Other Energy Storage | 0 | 0 | OK |
| Net Generation (MW) from Other Fuel Sources | 87,473 | 87,473 | OK |
| Net Generation (MW) from Pumped Storage | 10,032 | 10,032 | OK |
| Net Generation (MW) from Solar | 21,807 | 21,807 | OK |
| Net Generation (MW) from Solar with Integrated Battery Storage | 10,080 | 10,080 | OK |
| Net Generation (MW) from Solar without Integrated Battery Storage | 65,666 | 65,666 | OK |
| Net Generation (MW) from Unknown Energy Storage | 9,985 | 9,985 | OK |
| Net Generation (MW) from Unknown Fuel Sources | 0 | 0 | OK |
| Net Generation (MW) from Wind | 21,807 | 21,807 | OK |
| Net Generation (MW) from Wind with Integrated Battery Storage | 0 | 0 | OK |
| Net Generation (MW) from Wind without Integrated Battery Storage | 65,666 | 65,666 | OK |

## 10. Hand-off and evidence

**To the fact-load owner (Muaz):** `stg_eia930_balance_clean` (87,720 rows; business keys authority_code, data_date, hour_number / local_hour_end, utc_hour_end, source_file + source_sha256, source_row). Unpivot the 19 `gen_*` columns using `stg_energy_source_mapping`, whose `source_column` holds the exact original header and `clean_column` the matching staging column. Only non-NULL values become generation facts.

**To the dimension-load owner (Emaad):** `stg_authority_reference`, `stg_energy_source_mapping` and `stg_source_file` carry the natural keys (authority code, exact fuel header, file name + SHA-256).

**To the quality owner (Mohid):** missing-value and cleaning evidence is in `stg_cleaning_log`, `stg_load_exception` and the 30 PASS / FAIL checks in `sql/01_staging/90_verify_staging.sql`.

# 11. Reproducing the results

| File | Purpose |
|---|---|
| `sql/01_staging/00_create_staging_tables.sql` | Creates project_staging tables and the safe-parse helper functions |
| `etl/01_load_staging.py` | Loads references, manifest and the four CSVs (COPY), reconciles counts, runs the cleaning |
| `sql/01_staging/10_clean_staging.sql` | Cleaning rules C01-C11; rebuilds clean table, exceptions and log in one transaction |
| `sql/01_staging/90_verify_staging.sql` | 30 PASS / FAIL checks plus evidence queries |
| `sql/01_staging/95_ and 96_ selftest files` | Insert / remove the nine dirty rows (used by --self-test) |
| `reports/phase2/build_data_cleaning_report.py` | Regenerates this PDF from the database |

All scripts are re-runnable: running them again rebuilds identical counts and changes no results.